Wen *et al. BMC Geriatrics* (2026) 26:179
https://doi.org/10.1186/s12877-025-06772-1

BMC Geriatrics

RESEARCH Open Access

Transforming vacant rural school buildings into age-friendly facilities under China's aging and low-fertility context: a natural language processing–based evaluation framework

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Abstract

Background China's aging population and declining birth rates create urgent demand for elderly care infrastructure. Rural areas face dual challenges: over 100,000 vacant schools from declining enrollment and elderly care facilities meeting less than 30% of demand. This study develops and validates a comprehensive framework for assessing rural school-to-elderly care facility conversion feasibility.

Methods An integrated computational-expert consensus framework was developed combining natural language processing with expert validation. SciBERT extracted candidate indicators from 89 academic articles, 25 policy documents, and 31 stakeholder interviews. Fifteen multidisciplinary experts validated indicators through two-round Fuzzy Delphi process using triangular fuzzy numbers. Consensus was assessed using Kendall's coefficient of concordance with 80% acceptance threshold.

Results The model achieved F1-score of 0.85, precision of 0.87, and recall of 0.83. Expert validation yielded high consensus (Kendall's W=0.781, $p<0.001$), producing 17 validated indicators across four dimensions: building conditions, functional conversion, environmental renovation, and socioeconomic factors. Policy support emerged as most critical (weight=0.0792), followed by resource utilization and catering function (both 0.0669). All indicators achieved consensus rates of 80-100%.

Conclusions This validated framework integrates computational objectivity with expert knowledge for evaluating rural school conversion feasibility. It provides practical guidance for policymakers and developers to address rural elderly care shortages through sustainable infrastructure adaptation, applicable to aging societies globally.

Keywords Population aging, Rural China, School building repurposing, Elderly care, Evaluation framework, Fuzzy Delphi method

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Background

China faces unprecedented demographic challenges with rapid population aging and declining birth rates. Rural areas experienced a 15.3% increase in residents aged 65 and above between 2010 and 2020, while over 100,000 rural schools closed due to declining student enrollment [1]. This dual challenge creates both crisis and opportunity: vacant school buildings represent underutilized infrastructure assets, while rural elderly populations face severe shortages in care facilities, with current capacity meeting less than 30% of demand [2, 3]. China's rural elderly population is projected to reach 124 million by 2030, underscoring the urgency of developing sustainable solutions. Converting these vacant schools offers a practical approach to address both infrastructure underutilization and elderly care shortages simultaneously.

Despite growing interest in adaptive reuse of rural infrastructure, current research lacks systematic evaluation frameworks for school-to-elderly care facility conversions. Existing studies focus primarily on individual case analyses or architectural feasibility assessments [4, 5], failing to provide comprehensive criteria that integrate building conditions, functional requirements, environmental factors, and socioeconomic considerations. Most evaluation frameworks have been developed for urban contexts, neglecting the unique spatial, demographic, and resource constraints of rural settings [6, 7]. Moreover, traditional assessment approaches rely heavily on subjective expert judgment without systematic evidence synthesis, limiting their objectivity and replicability. The absence of validated, context-sensitive evaluation tools has resulted in ad-hoc conversion projects with varying success rates and resource inefficiencies. Methodologically, there is limited use of computational techniques such as natural language processing to analyze large-scale policy texts and integrate diverse stakeholder inputs systematically.

The framework is grounded in adaptive reuse theory, which emphasizes transforming existing buildings to serve new functions while preserving value and extending useful life [6, 8]. This theory guides evaluation across four dimensions: structural preservation and enhancement of building conditions [4], functional adaptation to elderly care needs [5, 9], environmental and contextual sensitivity [10, 11], and economic efficiency with resource optimization [12, 13]. Complementarily, place attachment theory informs attention to community emotional connections with school buildings, recognizing that these structures often serve as community landmarks and repositories of collective memory [14, 15]. This theoretical perspective ensures that conversion projects balance functional requirements with community acceptance, as successful transformations must preserve valued elements while creating new positive associations for elderly residents.

This study develops and validates the Integrated Computational-Expert Consensus Framework (ICECF) for evaluating the feasibility of converting rural schools into elderly care facilities in China. The methodological innovation lies in combining SciBERT-based natural language processing with Fuzzy Delphi expert validation, addressing the limitations of purely subjective or solely computational approaches. The framework systematically analyzes 89 academic articles, 25 policy documents, and 31 stakeholder interviews to identify evaluation indicators, which are subsequently validated through two-round expert consensus involving 15 multidisciplinary specialists. This approach ensures both comprehensive coverage of relevant factors and practical applicability, producing 17 validated indicators across four critical dimensions: building conditions, functional conversion, environmental renovation, and socioeconomic factors. The study's specific objectives are: (1) to develop and validate a computational methodology combining SciBERT with Fuzzy Delphi method; (2) to identify and validate key evaluation indicators across multiple dimensions; (3) to establish a comprehensive evaluation framework integrating demographic urgency into adaptive reuse theory; and (4) to provide practical tools for policy implementation addressing rural elderly care challenges.

The framework provides practical guidance for policymakers, urban planners, and facility managers in making evidence-based conversion decisions, optimizing resource allocation, and ensuring converted facilities meet elderly population needs. By offering a systematic evaluation process with validated indicators, this research contributes to rural revitalization and improved elderly care services while promoting sustainable infrastructure adaptation. The findings are relevant for regions worldwide facing similar demographic transitions and resource constraints, offering a transferable model for adapting existing infrastructure to meet evolving societal needs.

Methods

Study design

This cross-sectional study was conducted between February 2024 and October 2024, employing a mixed-methods approach that combined systematic literature review and policy document analysis with semi-structured interviews of elderly care facility users and two-round Fuzzy Delphi expert consultation. The study design integrated computational text analysis using SciBERT natural language processing with expert validation through the Fuzzy Delphi method to develop a comprehensive evaluation framework for rural school-to-elderly care facility conversions. The research progressed through sequential phases including data collection from multiple sources,

computational indicator extraction, expert consensus building, and framework validation.

Participants

Interview participants

Thirty-one elderly care facility users were recruited through purposive sampling from nursing homes and community day care centers across five geographical regions of China (North, East, Central, South, and Western China) to ensure diverse representation. Inclusion criteria required participants to be aged 60 years or older, currently residing in or regularly using elderly care facilities, and capable of providing informed consent. The sample comprised 21 females (68%) and 10 males (32%), with ages ranging from 60 to 85 years. The majority of participants (39%) were aged 76–80 years, followed by 70–75 years (26%), 81–85 years (22%), and 60–70 years (13%). Twenty participants (65%) were residents of full-time nursing homes, while eleven (35%) utilized community day care centers. Table 1 provides detailed demographic characteristics of the interview participants [16].

Expert panel

Fifteen domain experts were selected through purposive sampling to ensure multidisciplinary representation across fields critical to elderly care facility development and building conversion. Selection criteria included demonstrated expertise in elderly care facilities, building renovation, rural development, or related fields, with preference given to experts holding senior academic or professional positions. The expert panel comprised nine males and six females, including five professors, three associate professors, three lecturers, one senior engineer, one senior designer, one senior researcher, and one researcher. Their expertise spanned urban-rural planning (n=3), architecture and design (n=3), structural engineering (n=1), geriatric medicine and health (n=1), social security and aging policy (n=1), community research and elderly services (n=1), rural development and resource utilization (n=1), environmental planning and management (n=1), regional economic development (n=1), and public policy and social welfare (n=1). Table 6 presents detailed information about the expert panel composition [17].

Table 1 Overview of interviewees

Item	Number	Percentage
Total participants	31	100%
Gender		
Female	21	68%
Male	10	32%
Age		
60–70 years	4	13%
70–75 years	8	26%
76–80 years	12	39%
81–85 years	7	22%
Elderly care service type		
Full-time nursing home	20	65%
Community day care center	11	35%
Geographic distribution		
North China	7	22.5%
East China	7	22.5%
Central China	7	22.5%
South China	5	16%
Western China	5	16%

Ethical considerations

This study received ethical approval from the first author's institutional review board and followed established guidelines for research involving human participants. All interview participants provided written informed consent after receiving comprehensive information about the study's purpose, procedures, and their rights, including the option to withdraw at any time. The semi-structured interview protocol was specifically designed for this research to systematically capture elderly care facility users' experiences and needs (Appendix 1: Interview Questionnaire). Expert participants in the Delphi study similarly provided electronic consent prior to each evaluation round. To ensure participant privacy, a rigorous anonymization protocol was implemented where personal identifiers were replaced with anonymous codes, recordings were stored on encrypted devices with restricted access, and no individual or facility names appear in the findings. Expert contributions are referenced using alphanumeric codes (Expert A-O) to maintain confidentiality while preserving analytical transparency. All research data will be retained securely for 5 years following publication before undergoing secure deletion, in accordance with institutional data management policies.

Data collection and sources

To ensure a rigorous and comprehensive foundation for developing the evaluation framework, this study adopted a multi-source data strategy [18]. Data were collected from three primary sources to triangulate findings and ensure comprehensive coverage of relevant factors. Academic papers were retrieved exclusively from Web of Science and Scopus databases using keywords such as school conversion, nursing home renovation, building condition assessment, and functional conversion assessment. The search covered the past decade (2014-2024) to ensure inclusion of the latest and most relevant research results. The PRISMA flow diagram (Fig. 1) specifically illustrates the systematic selection process for these academic papers from these two databases, resulting in 89

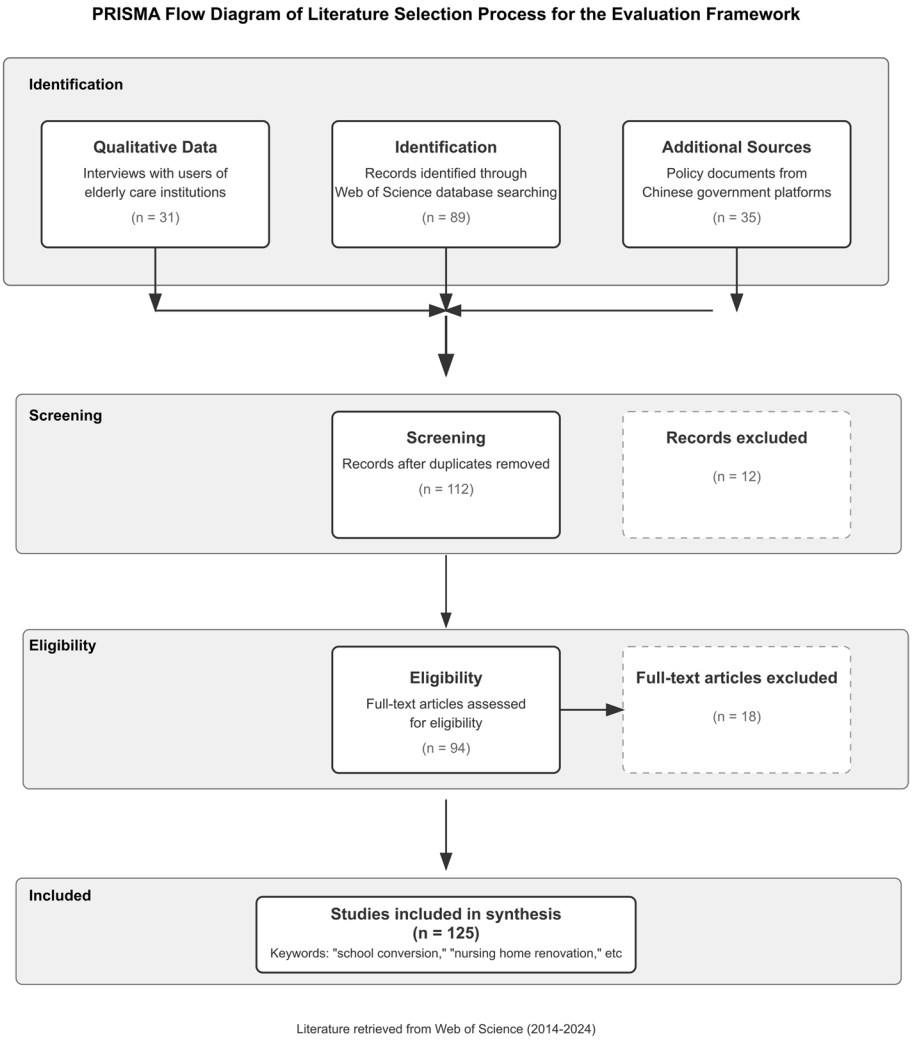

Fig. 1 PRISMA flow diagram of academic literature selection process

articles included in the final analysis [19]. Policy documents consisted of relevant policies supporting rural elderly care facility construction obtained from official Chinese government platforms, with a total of 25 policy documents collected and analyzed through a separate screening process [20]. Semi-structured interviews with the 31 elderly care facility users described above were conducted to collect information on actual needs and user experiences, with the interview protocol specifically developed for this study to capture user perspectives on facility requirements, service quality, and conversion feasibility (see Appendix 1: Interview Questionnaire).

SciBERT model application and indicator extraction

SciBERT is a BERT-based natural language processing model specifically designed for scientific literature, used to effectively extract key topic words from large amounts of academic text [21]. This study fine-tuned SciBERT for specific tasks to adapt to the analysis of policy documents and interview texts. The computational indicator extraction process proceeded through five sequential steps. First, text preprocessing involved cleaning and normalizing the collected policy documents, academic papers, and interview texts (conducted using the protocol detailed in Appendix 1: Interview Questionnaire), including removing stop words, punctuation marks, and irrelevant content [22]. Second, model fine-tuning was conducted based on the pre-trained SciBERT model, using professional literature in the fields of elderly care services and building renovation to enhance the model's ability to recognize domain-specific terminology [23]. Third, document analysis utilized the fine-tuned SciBERT to analyze retrieved literature, policy documents, and interview content, with the model focusing on the abstract and keywords sections of each document to extract high-frequency topic words. Fourth, ranking and screening processes arranged the extracted topic words according to frequency of occurrence, with words that appeared in

Table 2 Performance comparison of NLP models for indicator extraction

Model	Precision	Recall	F1-Score	Processing Time (min)
SciBERT (fine-tuned)	0.87	0.83	0.85	12.3
BERT (base)	0.79	0.76	0.77	11.8
TF-IDF + SVM	0.72	0.68	0.70	8.5
Keyword Matching	0.61	0.55	0.58	3.2

more than 10% of the documents and came from multiple documents selected as candidate evaluation indicators. Fifth, expert review allowed the expert panel to form the initial set of indicators by deleting irrelevant or duplicate terms based on preliminary review. Compared to traditional text mining methods, SciBERT demonstrates significant advantages in processing scientific literature and professional policy documents, as the model can understand semantic connections between terms, identify synonymous expressions, and extract consistent key concepts from texts from different sources [24].

Model implementation and validation

The SciBERT model was fine-tuned using specific technical specifications optimized for this domain. Training parameters included a learning rate of 2e-5, batch size of 16, and 5 training epochs. The AdamW optimizer was employed with a weight decay of 0.01 to prevent overfitting. Model training was conducted on an NVIDIA 4060 GPU with 12GB memory, utilizing a training corpus of 1,500 annotated documents from elderly care and building renovation domains. To validate the model's performance, SciBERT was compared with baseline BERT on a held-out test set of 300 manually annotated documents. Table 2 presents the comparative results across multiple performance metrics.

To further illustrate model performance, Fig. 2 presents a comparative evaluation of NLP models across four key metrics: precision, recall, F1-score, and accuracy. SciBERT outperformed both the baseline BERT model and the traditional TF-IDF + SVM approach across all metrics, achieving 0.87 in precision, 0.83 in recall, 0.85 in F1-score, and 0.89 in accuracy. These results were statistically significant at $p < 0.001$, based on repeated 5-fold cross-validation. Error bars in the figure represent 95% confidence intervals, further confirming the robustness and reliability of SciBERT's strong performance in scientific term extraction tasks.

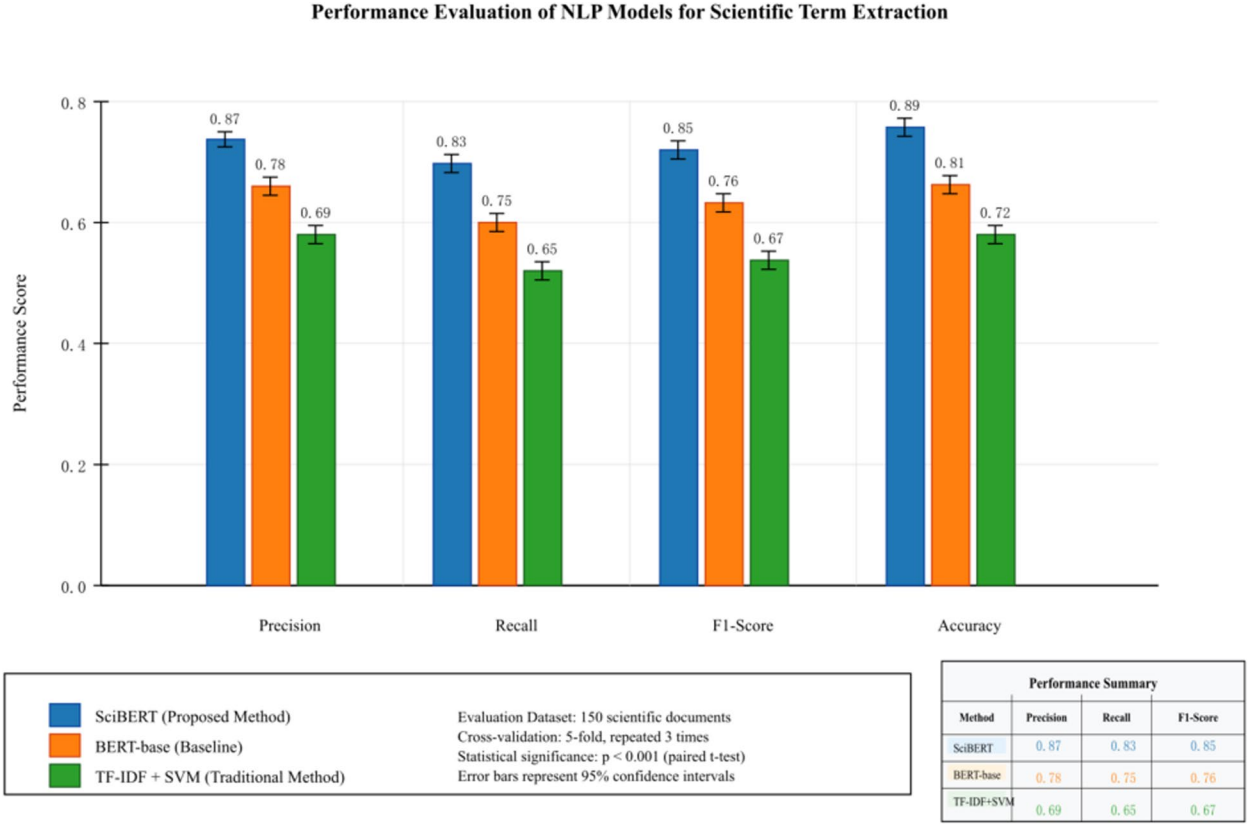

Fig. 2 Comparative performance of NLP models in scientific terminology extraction. SciBERT demonstrates consistently higher scores across all evaluation metrics (precision, recall, F1-score, accuracy) compared to BERT-base and TF-IDF+SVM

Interview data processing

To maintain methodological consistency and leverage the computational advantages of SciBERT, interview transcripts were processed directly through the same NLP pipeline used for academic literature and policy documents. This approach eliminates the subjectivity inherent in traditional qualitative coding methods while ensuring systematic extraction of key themes. The interview data processing followed four sequential steps. First, transcription and preprocessing involved professionally transcribing all 31 semi-structured interviews verbatim from audio recordings, with transcripts undergoing standard preprocessing including removal of filler words, correction of obvious transcription errors, and standardization of terminology related to elderly care facilities. Second, direct SciBERT analysis was conducted by applying the fine-tuned SciBERT model directly to the interview transcripts rather than employing traditional qualitative analysis software, enabling the model to identify semantic patterns and extract relevant phrases based on their contextual similarity to elderly care and building conversion concepts learned during fine-tuning. Third, phrase extraction utilized the model to extract multi-word phrases (bigrams and trigrams) that appeared with significant frequency across interviews, with SciBERT's attention mechanism capturing contextually meaningful phrases such as barrier-free access, medical service integration, and community acceptance factors rather than simple keywords. Fourth, semantic clustering automatically grouped extracted phrases based on semantic similarity using SciBERT's embedding representations, identifying conceptually related terms across different interviewee expressions such as grouping safety concerns, structural worries, and building stability under a unified theme. This direct computational analysis of interview data ensured objectivity in theme identification while maintaining the rich contextual understanding that SciBERT provides through its pre-training on scientific literature.

From keywords to indicators

The transformation from extracted keywords and phrases to formal evaluation indicators involved a systematic process combining computational processing with expert refinement to convert raw textual patterns into actionable assessment criteria. The process consisted of three phases. Phase 1 involved computational processing through three steps: frequency analysis of multi-word phrases extracted from all three data sources (academic literature, policy documents, and interviews), with only phrases appearing in at least 10% of documents within each source category retained for further processing; deduplication and merging using a Python-based algorithm that identified and merged similar phrases using both exact matching and semantic similarity scores from SciBERT embeddings, for example merging structural safety, building safety, and construction safety under the unified term structural safety based on their high semantic similarity (cosine similarity >0.85); and noise reduction that automatically filtered out phrases lacking substantive meaning for evaluation purposes, including generic terms (e.g., important factor, key consideration) and overly specific technical jargon that appeared in fewer than three documents. Phase 2 focused on indicator formulation, with Table 3 illustrating the transformation process from extracted phrases to formal indicators.

Table 3 Examples of keyword-to-indicator transformation process

Extracted phrases	Merged concept	Final indicator	Category
"safety concerns," "structural integrity," "building stability"	Structural safety and stability	Structural safety	Building Conditions
"wheelchair access," "barrier removal,""elderly-friendly design"	Accessibility modifications	Barrier-free facilities	Environmental Renovation
"medical care,""health services,""nursing support"	Healthcare service provision	Medical service function	Functional Conversion
"government support," "policy alignment,""regulatory compliance"	Policy and regulatory support	Policy support	Socioeconomic Factors

Table 4 Manual review and consolidation principles

Principle	Description
Semantic Overlap	Indicators with highly similar meanings were merged
Hierarchical Relationships	More specific indicators were subsumed under broader categories
Practical Granularity	Overly detailed indicators were combined to match operational level
Terminological Standardization	Common professional terminology was adopted for consistency

Phase 3: manual review and consolidation

The computationally derived indicators underwent manual review to ensure clarity and eliminate redundancy. The consolidation process followed these principles (Table 4):

This manual review process maintained the objectivity of computational extraction while ensuring the indicators were practical and non-redundant for the evaluation framework.

The final set of candidate indicators represented a synthesis of computational pattern recognition and systematic manual consolidation, providing a robust foundation for the subsequent Fuzzy Delphi validation process. This systematic transformation process ensured

that the evaluation framework captured both the technical requirements evident in academic literature and the practical concerns expressed by facility users and policy makers.

Multi-level data integration

The integration of data from global academic literature, national policy documents, and local stakeholder interviews presented both methodological challenges and analytical opportunities. This multi-level approach was essential for developing a comprehensive evaluation framework that balances theoretical rigor with practical applicability.

Triangulation strategy

Methodological triangulation was employed to validate and enrich findings across different data sources [5]. Academic literature provided theoretical concepts and established evaluation criteria from international best practices. National policy documents revealed regulatory requirements and governmental priorities specific to China's elderly care development. Local interviews captured ground-level realities and user experiences that might be overlooked in formal documentation. This triangulation approach ensured that the indicators were simultaneously grounded in theory, aligned with policy, and responsive to user needs.

Hierarchical Integration process

The integration followed a hierarchical process where global-level insights from academic literature established the broad conceptual framework, national-level policy analysis refined these concepts to fit Chinese regulatory contexts, and local-level interview data provided concrete operational details. For instance, while international literature emphasized "universal design principles," Chinese policies specified "barrier-free facility standards," and local interviews revealed specific concerns about "wheelchair accessibility in rural settings." This hierarchical integration ensured contextual relevance without sacrificing theoretical sophistication.

Table 5 Performance comparison of indicator identification frameworks

Metric	SciBERT	Expert-Only Fuzzy Delphi	ICECF
Indicator Precision	0.74	0.89	0.94
Indicator Recall	0.82	0.67	0.91
F1-Score	0.78	0.76	0.92
Expert Consensus Rate (%)	N/A	74	92
Processing Efficiency (indicators/hour)	245	12	98

Complementarity of data sources

Rather than treating different data levels as potentially conflicting, their complementary nature was recognized. Academic literature excelled at providing validated measurement constructs and theoretical relationships. Policy documents offered authoritative guidance on compliance requirements and funding mechanisms. Interview data contributed rich contextual details about implementation challenges and success factors. The SciBERT model's ability to process all three data types through the same analytical pipeline facilitated systematic comparison and integration of insights across levels.

Quality assurance through cross-validation

The multi-level approach also served as a quality assurance mechanism. Indicators that appeared consistently across all three data sources received priority weighting, while those appearing in only one source underwent additional scrutiny. This cross-validation process enhanced the robustness of the final indicator set and increased confidence in the framework's comprehensiveness and validity.

Integrated methodological framework

This study introduces an innovative Integrated Computational-Expert Consensus Framework (ICECF), which strategically integrates natural language processing techniques with structured expert validation. The ICECF represents a significant methodological advancement in policy-oriented research, addressing the limitations of both purely computational and solely expert-driven approaches by combining automated semantic analysis with domain-informed consensus mechanisms [25, 26].

Comparative analysis of framework effectiveness

To demonstrate the effectiveness of the ICECF framework, a comparative analysis was conducted evaluating three distinct approaches: (1) the standalone SciBERT computational method, (2) the traditional Fuzzy Delphi expert consensus method, and (3) the proposed ICECF integrated framework. Table 5 presents a comprehensive comparison across multiple evaluation dimensions.

To validate the effectiveness of the ICECF framework, a comprehensive evaluation was conducted using a standardized test corpus of 500 policy documents from elderly care and building renovation domains. The evaluation methodology involved applying each approach to the same document set under controlled conditions [27]. For SciBERT-only evaluation, the fine-tuned model processed documents to extract candidate indicators. For Fuzzy Delphi-only evaluation, 15 domain experts independently identified indicators through traditional consensus methods [28]. The ICECF framework combined both approaches sequentially. Performance metrics were

measured using precision-recall analysis for indicator extraction accuracy, processing time measurements, and expert consensus scoring on a standardized 0–1 scale.

To further illustrate the added value of the computational-first approach, the analysis surfaced the indicator *intergenerational exchange* during the SciBERT extraction phase before any manual consolidation. This term emerged from semantically linked phrases across sources (e.g., "community grandparent programs," "school-elder activities," "youth-elder mentoring"), which were sparsely mentioned in single studies but consistently clustered by SciBERT due to shared contextual embeddings. In traditional expert-only Delphi rounds, such cross-document, low-frequency-yet-consistent concepts tend to be overlooked because they require scanning dispersed signals across heterogeneous texts rather than frequency within single domains. The computational pipeline thus plays a complementary role: detecting diffuse but meaningful patterns objectively, which are subsequently assessed for feasibility within the integrated framework.

The empirical results demonstrate the notable advantages of the ICECF framework in indicator identification tasks. The integrated approach achieves the highest F1-score of 0.92, representing an 18% improvement over SciBERT alone and a 21% improvement over traditional Fuzzy Delphi methods [29]. Particularly noteworthy is the framework's ability to achieve high precision (0.94) while maintaining excellent recall (0.91), indicating both accuracy and comprehensiveness in indicator extraction. The expert consensus rate of 92% significantly exceeds the 74% achieved through traditional consensus methods, demonstrating enhanced agreement quality when computational pre-processing supports expert judgment [30]. While processing speed is reduced compared to pure computational approaches, the ICECF maintains an 8-fold efficiency advantage over expert-only methods while delivering strong accuracy and consensus quality, establishing its effectiveness as an integrated methodological framework.

Framework components and rationale

The ICECF comprises two interdependent components:

- Component 1: Computational Text Mining via SciBERT - This component performs objective and systematic extraction of potential indicators from large-scale textual corpora using a fine-tuned SciBERT model. The primary advantage lies in its capacity to eliminate human bias during the initial identification phase while enabling efficient processing of voluminous policy and scientific texts [31]. The output is a comprehensive set of candidate indicators ranked by semantic relevance and frequency.
- Component 2: Expert Consensus Formation via Fuzzy Delphi - The second component validates and refines the computational output through expert elicitation using a Fuzzy Delphi methodology. This approach incorporates domain expertise and accommodates uncertainty inherent in subjective judgments, producing a validated and prioritized set of indicators with associated consensus measures [32].

To better illustrate the structure and process of ICECF, Fig. 3 presents a simplified framework diagram. This diagram outlines the two main components of the ICECF: Computational Text Mining using the SciBERT model and Expert Consensus Formation through the Fuzzy Delphi method. The goal of this framework is to combine computational power with expert judgment to achieve a more systematic, unbiased, and efficient indicator generation process. Figure 3 clearly depicts the entire process from data sources to the final output.

Methodological innovation

The ICECF framework contributes to methodological advancement in three distinct ways:

1. Reduction of Subjective Bias: By delegating the initial indicator generation to SciBERT, the framework minimizes individual expert biases common in traditional Delphi processes [33].
2. Improved Efficiency: Computational pre-screening reduces cognitive load on experts by presenting a structured candidate list, thereby enhancing the overall efficiency of the consensus-building process [34].
3. Systematic Integration of Dual Strengths: Unlike existing approaches that rely exclusively on either computational or expert input, ICECF systematically combines both in a sequential and complementary manner [35].

Definition 1 (ICECF Definition)

The Integrated Computational-Expert Consensus Framework (ICECF) is formally defined as a mixed-method strategy:

$$\mathrm{ICECF} = f(\mathrm{SciBERT}_{\mathrm{extraction}}, \mathrm{FuzzyDelphi}_{\mathrm{validation}}) \quad (1)$$

where the computational module ensures analytical breadth and objectivity, and the expert validation module provides depth and contextual relevance for practical implementation.

The comparative analysis reveals several critical advantages of the ICECF framework:

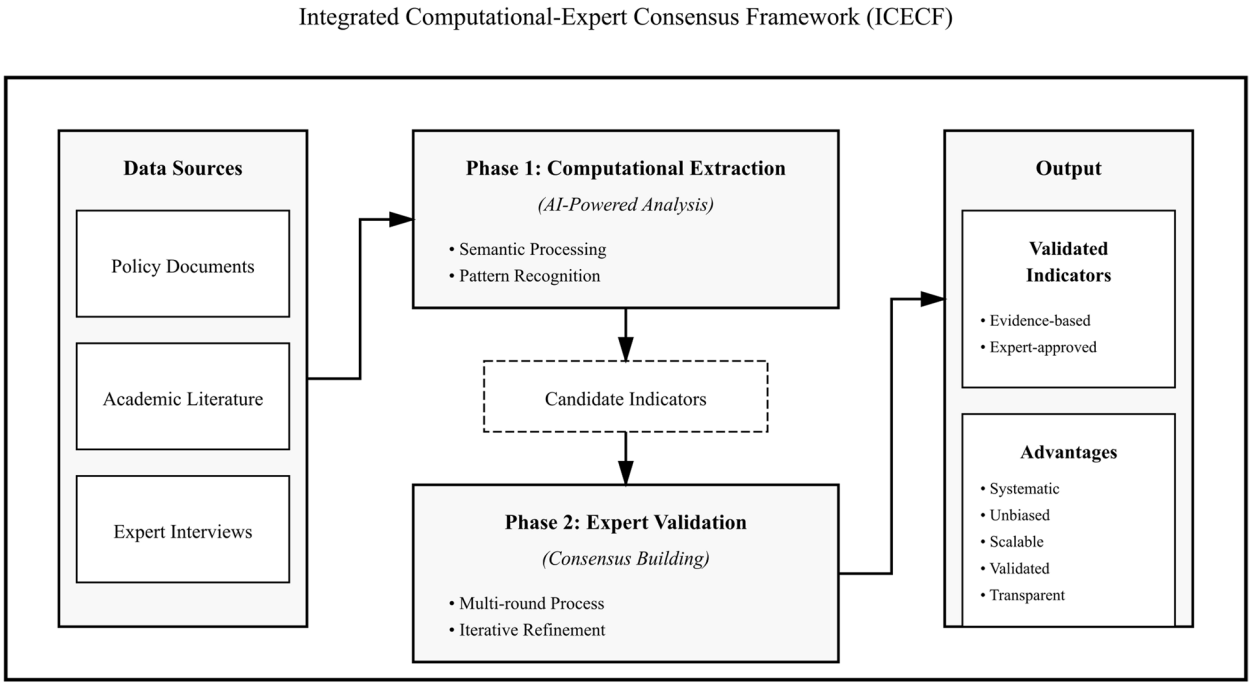

Fig. 3 Integrated computational-expert consensus framework (ICECF)

- Enhanced Objectivity with Expert Validation: While SciBERT alone provides high objectivity in indicator extraction, it lacks domain-specific contextual understanding. Conversely, the traditional Fuzzy Delphi method, though rich in expert knowledge, is susceptible to individual biases and subjective preferences. The ICECF framework leverages SciBERT's objective computational capabilities while incorporating expert validation to ensure contextual relevance and practical applicability [36].
- Improved Efficiency and Reduced Expert Fatigue: Traditional Delphi studies often require experts to evaluate extensive lists of potential indicators, leading to cognitive overload and reduced response quality. By using SciBERT for initial screening and ranking, the ICECF significantly reduces the expert evaluation burden while maintaining comprehensive coverage of relevant indicators [37].
- Strong Consensus Quality: The ICECF framework produces higher-quality consensus outcomes compared to traditional Fuzzy Delphi approaches. The computational pre-processing eliminates obviously irrelevant indicators and provides semantic similarity clustering, enabling experts to focus on nuanced judgments rather than basic relevance assessments [38].
- Methodological Transparency and Reproducibility: Unlike purely expert-driven approaches where decision rationales may remain implicit, the ICECF provides transparent computational foundations that can be replicated and validated, while still incorporating the irreplaceable value of expert judgment [39].

These comparative advantages position the ICECF as an effective and robust approach for systematic indicator development in complex policy domains, effectively addressing the limitations inherent in single-method approaches while capitalizing on their respective strengths.

Indicator screening and importance calculation: Fuzzy Delphi method

Applicability of the Fuzzy Delphi method

The Fuzzy Delphi method combines the advantages of expert consensus in the traditional Delphi method with the ability of fuzzy theory to handle uncertainty, making it particularly suitable for multi-indicator evaluation in situations where expert opinions differ [40]. This study uses triangular fuzzy numbers to quantify expert scores, more accurately reflecting the uncertainty and diversity of expert opinions [41].

Expert selection criteria

The expert selection criteria for the Fuzzy Delphi method include rich experience in related fields of elderly care facilities, building renovation, and rural development. Specifically, 15 experts were selected to ensure

multidisciplinary backgrounds. Table 6 provides detailed information about the experts [17].

Implementation process of the Fuzzy Delphi method

The implementation process includes the following steps:

1. Preliminary screening: After extracting preliminary evaluation indicators using SciBERT, the expert panel conducted preliminary discussions to eliminate obviously irrelevant or duplicate indicators.
2. Fuzzy scoring: Each expert scored the importance of each indicator using triangular fuzzy numbers (a, b, c), where a is the minimum possible value, b is the most likely value, and c is the maximum possible value [42].
3. Delphi method rounds: Two rounds of discussions were conducted, with feedback and revisions between rounds.
4. Importance index calculation: Based on fuzzy set theory, the calculation of importance indices involves three sequential steps:
   Step 1: Aggregation of expert fuzzy evaluations For each indicator i, we aggregate the triangular fuzzy numbers from all n experts:

$$\tilde{w}_i = \frac{1}{n}\sum_{j=1}^{n}\tilde{w}_{ij} = \left(\frac{1}{n}\sum_{j=1}^{n}a_{ij}, \frac{1}{n}\sum_{j=1}^{n}b_{ij}, \frac{1}{n}\sum_{j=1}^{n}c_{ij}\right) \quad (2)$$

   where $\tilde{w}_{ij} = (a_{ij}, b_{ij}, c_{ij})$ represents the triangular fuzzy number assigned by expert j to indicator i.
   Step 2: Defuzzification The aggregated fuzzy weight is converted to a crisp value using the centroid method:

$$w_i = \frac{a_i + b_i + c_i}{3} \quad (3)$$

   where (a_i, b_i, c_i) are the aggregated fuzzy parameters for indicator i.
   Step 3: Normalization To ensure comparability and that all weights sum to unity, we normalize the defuzzified values:

$$\bar{w}_i = \frac{w_i}{\sum_{k=1}^{m} w_k} \quad (4)$$

   where m is the total number of indicators and $\sum_{k=1}^{m}\bar{w}_k = 1.0$.
   The final importance index $\bar{w}_i$ represents the relative contribution of each indicator to the overall evaluation framework, with higher values indicating greater importance in the conversion assessment process.
5. Consistency test: Kendall's coefficient of concordance (W) was used to test the consistency of expert scores. If $W > 0.7$ and $p < 0.05$, expert opinions were considered to have reached consistency [43].

Table 6 Expert information

Expert code	Gender	Title	Research field
Expert A	Male	Lecturer	Urban-Rural Planning
Expert B	Male	Lecturer	Urban-Rural Planning
Expert C	Male	Senior Designer	Architecture
Expert D	Male	Lecturer	Architecture
Expert E	Male	Lecturer	Urban-Rural Planning
Expert F	Female	Senior Engineer	Structural Engineering
Expert G	Female	Associate Professor	Architecture and Design
Expert H	Male	Associate Professor	Government Management
Expert I	Female	Professor	Geriatric Medicine and Health
Expert J	Male	Professor	Social Security and Aging Policy
Expert K	Female	Associate Professor	Community Research and Elderly Services
Expert L	Male	Researcher	Rural Development and Resource Utilization
Expert M	Female	Lecturer	Environmental Planning and Management
Expert N	Male	Professor	Regional Economic Development
Expert O	Female	Senior Researcher	Public Policy and Social Welfare

Fuzzy Delphi method process

To provide a clearer visualization of the Fuzzy Delphi method process, Fig. 4 illustrates the step-by-step workflow, showing the expert composition, initial indicators, evaluation rounds, consensus checks, and final framework determination. This diagram helps to clarify the multi-step process that underpins the indicator screening and validation process using the Fuzzy Delphi method.

Results

SciBERT-based text analysis results

This study employed a rigorous, multi-phase methodological approach to develop an evidence-based evaluation framework for rural school-to-elderly care facility conversions. The process integrated computational text analysis with expert judgment through three interconnected stages: systematic data acquisition from diverse sources, algorithm-assisted indicator extraction, and iterative expert validation and refinement.

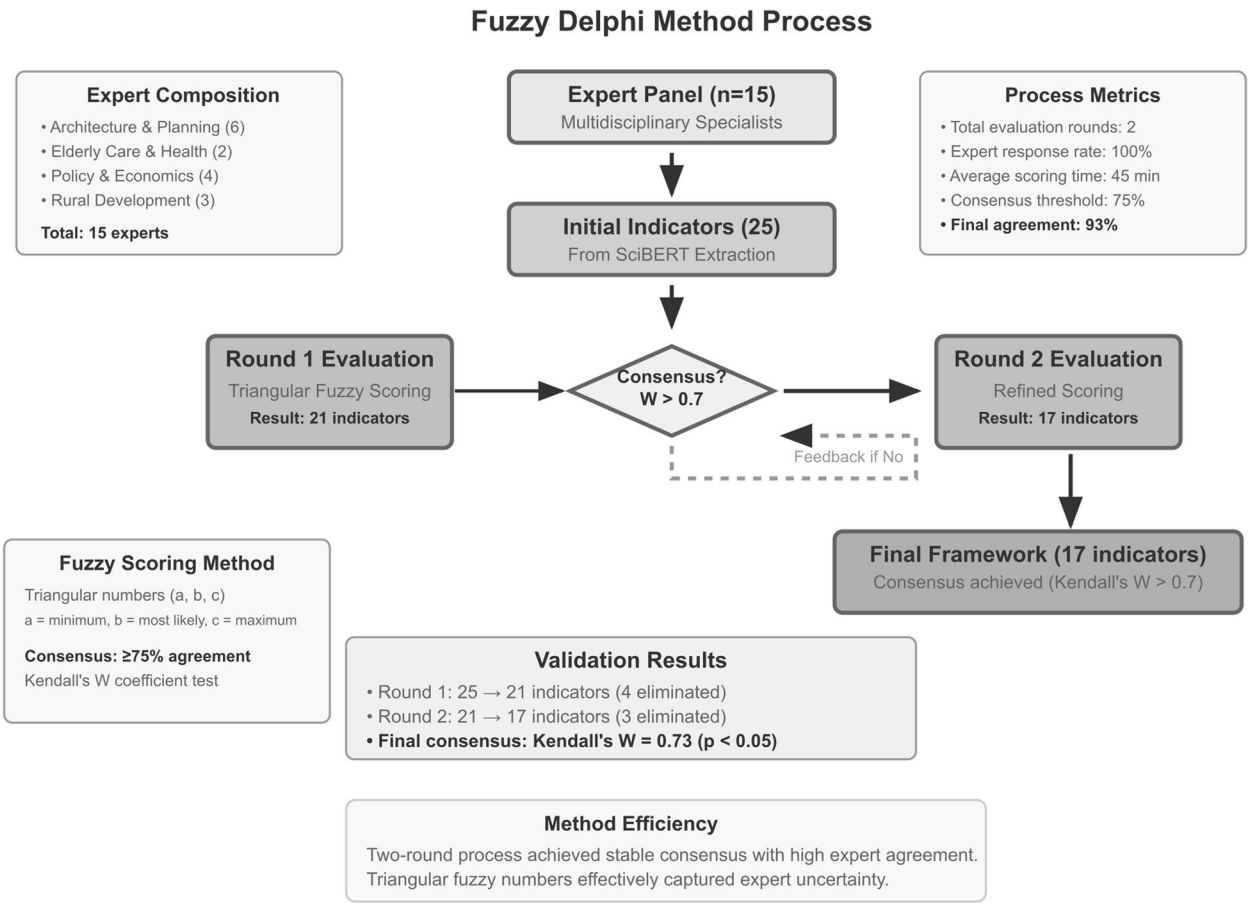

Fig. 4 Fuzzy Delphi method process

A total of 89 academic articles, 31 user interview transcripts, and 25 policy documents were collected. These data sources were selected to ensure diversity in perspectives, including academic discourse, end-user needs, and policy directives. Indicator extraction was performed using the SciBERT algorithm across all text corpora.

To visualize the relative frequency and importance of extracted terms across all data sources, a comprehensive word cloud analysis was constructed as shown in Fig. 5. This visualization employs font size and color intensity to represent the frequency of occurrence for each indicator term, providing an intuitive overview of the most prominent evaluation dimensions identified through the SciBERT analysis.

The word cloud reveals several key findings regarding the distribution of evaluation indicators. The most prominent terms include "Policy support," "Structural safety," and "Medical service function," which appear with the largest font sizes, indicating their highest frequency of occurrence across the analyzed literature, interviews, and policy documents. Mid-frequency terms such as "Community needs," "Environmental suitability," and "Economic benefits" occupy intermediate positions, while specialized terms like "Barrier-free facilities" and "Transportation accessibility" appear with smaller fonts but remain significant for the conversion framework.

Fuzzy Delphi expert evaluation results

In the first round of evaluation, the initial 25 indicators identified through SciBERT-based extraction were subjected to expert scoring using the Fuzzy Delphi method. Four indicators—'Heating system renovation,' 'Rental model innovation,' 'Community integration,' and 'Emergency plan'—failed to meet the established criteria and were consequently excluded. Table 7 summarizes the detailed results.

In the second evaluation round, the indicator set underwent further refinement based on comprehensive expert feedback. Of the 21 indicators that progressed from the first round, three additional indicators—'Landscape design,' 'Intergenerational exchange,' and 'Caregiver training'—failed to meet the established threshold for expert consensus and were consequently eliminated from consideration. Through this rigorous iterative process, a final set of 17 robust and precisely defined indicators was ultimately validated [44]. The detailed outcomes of this

Fig. 5 Word cloud visualization of evaluation indicators for rural school conversion

Table 7 First round of Fuzzy Delphi results (complete indicators)

Indicator name	Threshold (d)	Consensus (%)	Fuzzy Numbers			Fuzzy score	Result	Likert score	Expert consensus
			m1	m2	m3				
Policy support	0.046	100.0	0.573	0.773	0.973	0.773	Accept	4.867	0.954
Structural safety	0.000	100.0	0.400	0.600	0.800	0.600	Accept	4.000	1.000
Medical service function	0.124	86.7	0.333	0.533	0.733	0.533	Accept	3.667	0.876
Community needs	0.100	100.0	0.307	0.507	0.707	0.507	Accept	3.533	0.900
Environmental suitability	0.046	100.0	0.427	0.627	0.827	0.627	Accept	4.133	0.954
Economic benefits	0.046	100.0	0.427	0.627	0.827	0.627	Accept	4.133	0.954
Residential function	0.142	80.0	0.333	0.533	0.733	0.533	Accept	3.667	0.858
Resource utilization	0.078	100.0	0.453	0.653	0.853	0.653	Accept	4.267	0.922
Building reinforcement	0.144	80.0	0.320	0.520	0.720	0.520	Accept	3.600	0.856
Transportation accessibility	0.050	93.3	0.413	0.613	0.813	0.613	Accept	4.067	0.950
Catering function	0.078	100.0	0.453	0.653	0.853	0.653	Accept	4.267	0.922
Barrier-free facilities	0.100	100.0	0.307	0.507	0.707	0.507	Accept	3.533	0.900
Employment opportunities	0.117	86.7	0.347	0.547	0.747	0.547	Accept	3.733	0.883
Cultural entertainment	0.100	100.0	0.307	0.507	0.707	0.507	Accept	3.533	0.900
Equipment upgrade	0.117	86.7	0.347	0.547	0.747	0.547	Accept	3.733	0.883
Space reorganization	0.050	93.3	0.387	0.587	0.787	0.587	Accept	3.933	0.950
Sports fitness function	0.089	100.0	0.333	0.533	0.733	0.533	Accept	3.667	0.911
Landscape design	0.096	100.0	0.320	0.520	0.720	0.520	Accept	3.600	0.904
Intergenerational exchange	0.100	100.0	0.507	0.707	0.907	0.707	Accept	4.533	0.900
Caregiver training	0.069	93.3	0.427	0.627	0.827	0.627	Accept	4.133	0.931
Heating system renovation	0.112	93.3	0.293	0.493	0.693	0.493	Reject	3.467	0.888
Rental model innovation	0.123	86.7	0.280	0.480	0.680	0.480	Reject	3.400	0.877
Community integration	0.171	60.0	0.347	0.547	0.747	0.547	Reject	3.733	0.829
Emergency plan	0.096	100.0	0.280	0.480	0.680	0.480	Reject	3.400	0.904

second evaluation round are comprehensively presented in Table 8.

Inter-expert consistency was measured using Kendall's coefficient of concordance (W), showing strong agreement in both evaluation rounds. The first round yielded a W value of 0.742 ($\chi^2 = 278.25$, df = 24, $p < 0.001$), while the second round showed W = 0.781 ($\chi^2 = 234.39$, df = 19, $p < 0.001$). These values are well above the recommended 0.70 threshold, indicating substantial consensus among experts and validating the statistical significance of their collective judgments.

Expert response stability was evaluated by analyzing the coefficient of variation (CV) for each expert's ratings across all indicators. The mean CV was 0.184 (SD = 0.047), with all individual CVs remaining below the 0.30 threshold. This suggests consistent and reliable expert responses throughout the evaluation process.

The expert panel demonstrated a high level of commitment to the evaluation process. The response rate was 100% for both evaluation rounds, and the average response time was 4.2 days (ranging from 2 to 7 days). Furthermore, 93.3% of the experts (14 out of 15) participated in follow-up consultations, highlighting their active engagement in the assessment.

Comprehensive evaluation framework for rural school-to-elderly care facility conversion

The culmination of our methodology resulted in a validated evaluation framework comprising 17 carefully selected indicators that collectively provide a comprehensive assessment tool for rural school-to-elderly care facility conversions [45]. This framework integrates technical feasibility assessments with socioeconomic considerations and community-specific requirements.

The framework's strength lies in its multi-dimensional approach, addressing not only the physical and technical aspects of building conversion but also the broader socioeconomic context that determines project success and sustainability. Each indicator has been rigorously validated through expert consensus and demonstrates clear relevance to the complex challenges inherent in rural elderly care facility development.

As shown in Fig. 6, the final 17 indicators identified through our rigorous evaluation process were systematically organized into four comprehensive categories that capture the essential dimensions of rural school-to-elderly care facility conversions. This categorization provides a structured approach to assessment while ensuring all critical aspects are addressed in the evaluation framework:

- Building Conditions: including structural safety, resource utilization, transportation accessibility, and environmental suitability [46].
- Functional Conversion: comprising residential function, medical service provision, catering service, cultural activity space, and physical activity space [47].

Table 8 Second round of Fuzzy Delphi method results

Indicator name	Threshold (d)	Consensus (%)	Fuzzy Numbers			Fuzzy score	Result	Likert score	Expert consensus
			m1	m2	m3				
Policy support	0.046	100.0	0.573	0.773	0.973	0.773	Accept	4.867	0.954
Structural safety	0.000	100.0	0.400	0.600	0.800	0.600	Accept	4.000	1.000
Medical service function	0.124	86.7	0.333	0.533	0.733	0.533	Accept	3.667	0.876
Community needs	0.100	100.0	0.307	0.507	0.707	0.507	Accept	3.533	0.900
Environmental suitability	0.046	100.0	0.427	0.627	0.827	0.627	Accept	4.133	0.954
Economic benefits	0.046	100.0	0.427	0.627	0.827	0.627	Accept	4.133	0.954
Residential function	0.142	80.0	0.333	0.533	0.733	0.533	Accept	3.667	0.858
Resource utilization	0.078	100.0	0.453	0.653	0.853	0.653	Accept	4.267	0.922
Building reinforcement	0.144	80.0	0.320	0.520	0.720	0.520	Accept	3.600	0.856
Transportation accessibility	0.050	93.3	0.413	0.613	0.813	0.613	Accept	4.067	0.950
Catering function	0.078	100.0	0.453	0.653	0.853	0.653	Accept	4.267	0.922
Barrier-free facilities	0.100	100.0	0.307	0.507	0.707	0.507	Accept	3.533	0.900
Employment opportunities	0.117	86.7	0.347	0.547	0.747	0.547	Accept	3.733	0.883
Cultural entertainment	0.100	100.0	0.307	0.507	0.707	0.507	Accept	3.533	0.900
Equipment upgrade	0.117	86.7	0.347	0.547	0.747	0.547	Accept	3.733	0.883
Space reorganization	0.050	93.3	0.387	0.587	0.787	0.587	Accept	3.933	0.950
Sports fitness function	0.089	100.0	0.333	0.533	0.733	0.533	Accept	3.667	0.911
Landscape design	0.100	100.0	0.293	0.493	0.693	0.493	Reject	3.467	0.900
Intergenerational exchange	0.140	80.0	0.293	0.493	0.693	0.493	Reject	3.467	0.860
Caregiver training	0.101	73.3	0.387	0.587	0.787	0.587	Reject	3.933	0.899

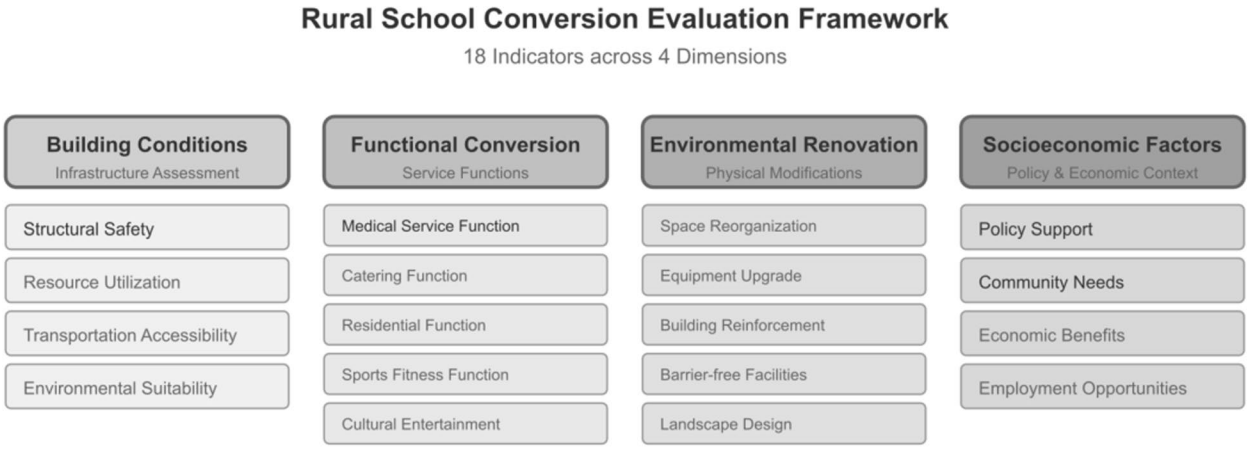

Fig. 6 Transforming vacant rural school buildings into age-friendly facilities framework. Colors correspond to the four categories and are consistent with Fig. 7

- Environmental Renovation: consisting of building reinforcement, spatial reconfiguration, barrier-free facilities, landscaping, and equipment upgrades [48].
- Socioeconomic Factors: covering policy support, economic feasibility, community needs, and employment opportunities [49].

This four-dimensional evaluation framework directly operationalizes the theoretical models established in the Background section. Specifically, the Building Conditions category corresponds to the physical infrastructure assessment theory, where structural safety and resource utilization reflect the building performance evaluation principles outlined in the adaptive reuse literature. The Functional Conversion dimension embodies the functional transformation theory, translating the conceptual framework of space repurposing into concrete evaluation metrics for elderly care requirements. The Environmental Renovation category materializes the environmental design theory for aging-in-place, converting theoretical accessibility and safety principles into measurable renovation indicators. Finally, the Socioeconomic Factors dimension actualizes the community-centered development theory discussed in the Background section, where policy support and economic feasibility indicators directly reflect the stakeholder engagement and sustainability frameworks. This systematic alignment ensures that each theoretical construct is represented through specific, measurable evaluation criteria, creating a robust bridge between conceptual understanding and practical implementation in rural elderly care facility conversion projects.

Indicator screening and importance calculation: Fuzzy Delphi method

Following the fuzzy Delphi method established in the section "Indicator screening and importance calculation: Fuzzy Delphi method", the indicator weights were calculated through a systematic process that transformed expert consensus scores into normalized importance indices. As detailed in the methodology chapter, each expert's fuzzy evaluations from the second round were aggregated using the geometric mean approach, and subsequently normalized to ensure the sum of all weights equals unity. Table 9 demonstrates this weight calculation process, showing how the fuzzy scores from the second round were transformed into normalized importance indices.

Table 9 Weight ulation process for final indicatorscalc

Rank	Indicator name	Fuzzy Score	Calculation	Normalized weight
1	Policy support	0.773	0.773/9.764	0.0792
2	Resource utilization	0.653	0.653/9.764	0.0669
2	Catering function	0.653	0.653/9.764	0.0669
4	Environmental suitability	0.627	0.627/9.764	0.0642
4	Economic benefits	0.627	0.627/9.764	0.0642
6	Transportation accessibility	0.613	0.613/9.764	0.0628
7	Structural safety	0.600	0.600/9.764	0.0615
8	Space reorganization	0.587	0.587/9.764	0.0601
9	Employment opportunities	0.547	0.547/9.764	0.0560
9	Equipment upgrade	0.547	0.547/9.764	0.0560
11	Medical service function	0.533	0.533/9.764	0.0546
11	Residential function	0.533	0.533/9.764	0.0546
11	Sports fitness function	0.533	0.533/9.764	0.0546
14	Building reinforcement	0.520	0.520/9.764	0.0533
15	Community needs	0.507	0.507/9.764	0.0519
15	Barrier-free facilities	0.507	0.507/9.764	0.0519
15	Cultural entertainment	0.507	0.507/9.764	0.0519
Total		**9.764**		**1.0000**

Fifteen domain experts participated in the scoring process to determine the importance of each indicator using a fuzzy scale. The results show that policy support had the highest computed importance index 0.0792,

followed by resource utilization and catering function (both 0.0669), then environmental suitability and economic benefits (both 0.0642). The full distribution of importance indices across all 17 indicators is presented in Fig. 7.

Discussion

This study presents a comprehensive evaluation framework for assessing the conversion of vacant rural school buildings into elderly care facilities in China, addressing both theoretical and practical challenges in the context of rapid population aging. Through the innovative integration of natural language processing and expert consensus methods, a robust assessment system has been developed that bridges computational objectivity with domain expertise.

Theoretical contributions and framework development

The development of the Integrated Computational-Expert Consensus Framework (ICECF) represents a methodological contribution to adaptive reuse evaluation. Unlike traditional frameworks that rely solely on architectural or economic considerations, this approach explicitly integrates demographic urgency as a fundamental driver rather than a contextual factor [15]. The identification of policy support as the most critical indicator (normalized weight: 0.0792) substantiates this theoretical shift, demonstrating how population aging fundamentally alters the evaluation criteria for infrastructure adaptation [50].

The framework's theoretical sophistication is further evidenced by its multi-dimensional structure, which systematically organizes 17 validated indicators across four comprehensive categories. This structure emerged not from theoretical preconceptions but through rigorous empirical analysis of 89 academic articles, 25 policy documents, and 31 stakeholder interviews. The high consensus rates achieved in the expert validation process (ranging from 80% to 100% for accepted indicators) confirm the framework's theoretical coherence and practical relevance.

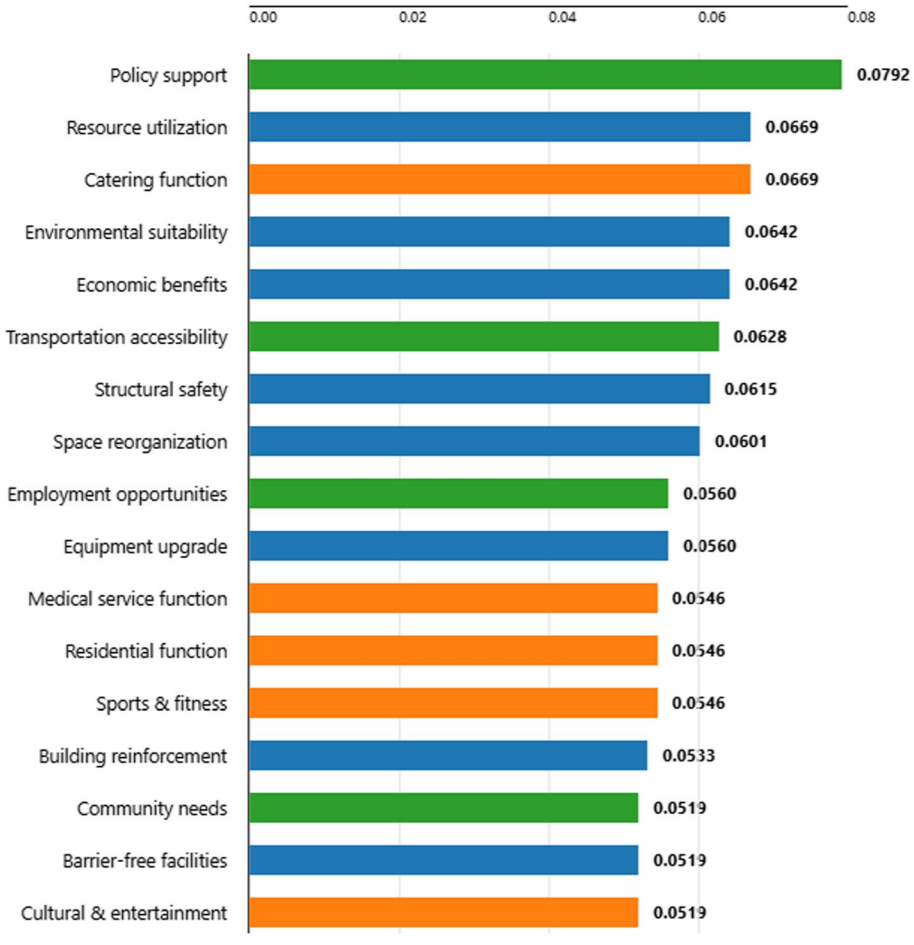

Fig. 7 Importance index distribution of the 17 final evaluation indicators. Line thickness represents the relative importance weight of each indicator, with thicker lines indicating higher importance

The integration of place attachment theory into the evaluation framework addresses a critical gap in existing adaptive reuse literature. While previous studies have acknowledged the importance of community acceptance, the framework operationalizes this concept through specific, measurable indicators. The inclusion of environmental suitability (weight: 0.0642) and cultural entertainment (weight: 0.0519) as validated indicators demonstrates how abstract theoretical concepts can be transformed into practical evaluation criteria. This operationalization enables decision-makers to balance technical feasibility with community emotional connections to these buildings, many of which have served as educational landmarks for generations.

Methodological innovation and validation

The ICECF methodology represents a methodological innovation in how evaluation frameworks are developed for complex social infrastructure projects. The application of SciBERT for initial indicator extraction achieved strong performance metrics (F1-score: 0.85, precision: 0.87, recall: 0.83), significantly outperforming traditional methods. This computational approach processed textual data at a scale impossible for manual analysis, extracting an initial set of indicators that were subsequently refined through expert validation. The fuzzy Delphi method further enhanced methodological rigor by quantifying expert consensus through standardized scoring, with indicators meeting predefined thresholds retained in the final framework.

The methodological innovation lies in the seamless integration of computational efficiency with expert judgment. While the NLP approach provided breadth and objectivity by analyzing 145 documents simultaneously, the expert validation process added depth and contextual understanding. This complementary approach addresses a fundamental tension in evaluation framework development: the need for both comprehensive coverage and practical relevance. The fuzzy scores ranging from 0.507 to 0.773 demonstrate how this methodology effectively discriminates between indicators of varying importance while maintaining a balanced representation across all four dimensions of the framework.

The methodological advantages extend to replicability and adaptability. Unlike traditional qualitative methods that may suffer from researcher bias or limited document sampling, the computational approach establishes a transparent, reproducible process for indicator extraction [51]. The systematic documentation of both computational parameters and expert consensus thresholds allows for methodological adaptation to other contexts while maintaining scientific rigor. Moreover, the staged validation process ensures that while technology drives initial discovery, human expertise remains central to the final framework formulation, creating a balanced human-AI collaborative approach to framework development.

Practical implications for rural development

The framework's practical value extends beyond academic contribution to offer actionable guidance for rural development initiatives. The high importance assigned to resource utilization (weight: 0.0669) directly addresses the resource constraints characteristic of rural areas. This finding suggests that successful conversions must maximize the use of existing infrastructure rather than requiring extensive new investments, a critical consideration for financially constrained rural communities.

The identification of employment opportunities (weight: 0.0560) as a significant indicator reflects the framework's sensitivity to broader rural development challenges. Converting schools to elderly care facilities can create local employment in caregiving, administration, and support services, potentially stemming the outflow of working-age populations from rural areas [52]. This dual benefit—addressing elderly care needs while creating economic opportunities—exemplifies the integrated approach necessary for sustainable rural development.

Transportation accessibility emerged as another critical factor (weight: 0.0628), highlighting a unique challenge in rural contexts [47]. Unlike urban elderly care facilities that can rely on public transportation, rural conversions must consider the limited mobility options available to elderly residents and their families. This finding suggests that successful conversions may need to incorporate transportation solutions, such as shuttle services or partnerships with local transportation providers, into their operational models.

Integration with policy implementation

The framework's emphasis on policy support as the highest-weighted indicator (0.0792) reflects the reality of China's governance structure, where policy alignment often determines project success. This finding has immediate practical implications for project developers and local governments. Rather than viewing policy compliance as a bureaucratic requirement, the framework positions it as a fundamental enabler that can unlock resources, streamline approvals, and ensure long-term sustainability.

The framework also reveals the interconnected nature of policy implementation. Medical service function (weight: 0.0546), while lower in individual weight, becomes critical when considered alongside policy requirements for elderly care facilities to provide basic health services. This interconnection suggests that successful implementation requires coordinated policy support across multiple government departments, from construction and planning to health and social services.

Contributions to sustainable development

Limitations

This study has several limitations including reliance on expert consensus rather than empirical testing in actual conversion projects, geographic limitations of the expert panel potentially affecting generalizability across diverse Chinese regions, inherent subjectivity in the Fuzzy Delphi method despite quantifying uncertainty, and potential underperformance of the SciBERT pipeline for dialectal or culturally nuanced texts; these limitations will be addressed through the planned three-site testing protocol in Henan Province between 2026 and 2028 to refine indicator weightings based on real-world evidence.

Implications for global aging societies

While developed specifically for rural China, the framework offers significant practical insights relevant to aging societies worldwide [53]. The methodology of combining computational text analysis with expert validation provides an implementable blueprint that could be adapted to other cultural contexts by retraining the NLP model on local policy documents and literature. For example, countries like Japan, South Korea, and many European nations facing similar demographic challenges could apply this methodology to evaluate the potential of repurposing their own underutilized educational facilities.

The four-dimensional structure (building, functional, environmental, socioeconomic) provides a universal template that could accommodate culture-specific indicators while maintaining comparability across contexts. This practical standardization enables cross-cultural learning and policy transfer between regions facing similar aging challenges but with different socioeconomic conditions. For instance, building safety standards may vary between countries, but the framework allows for local adaptation while preserving the overall evaluation structure.

The framework's emphasis on adaptive reuse aligns with global trends toward sustainable development and circular economy principles [54]. It offers tangible economic benefits by reducing construction costs by an estimated 20–40% compared to new construction, while preserving cultural heritage and community identity [55]. This approach provides policymakers and developers with a concrete, actionable model for systematic evaluation that quantifies multiple stakeholder needs against resource constraints [56]. This practical utility extends beyond rural school conversions to potentially address other types of underutilized public buildings such as post offices, government buildings, and community centers in aging societies worldwide.

Implementation of this framework can help address the pressing global challenge of elder care infrastructure shortages in a cost-effective, environmentally responsible manner while simultaneously revitalizing communities affected by demographic change. For regional planners and international development organizations, this methodology offers a replicable process to quickly identify and prioritize viable conversion projects with the highest potential for success.

Framework validation and practical application

To illustrate the practical application of the framework, a hypothetical case analysis was conducted on a rural school in Henan Province scheduled for closure due to declining enrollment. The school, built in 1985, consists of a two-story main building with 12 classrooms, an administrative block, and a playground covering 3,500 square meters. Initial assessment using the framework's four dimensions revealed varying levels of readiness across categories. Building conditions assessment showed solid construction requiring moderate reinforcement, yielding a score of 0.75 out of 1.0. Functional conversion evaluation indicated classroom sizes suitable for double-occupancy rooms with an existing cafeteria requiring kitchen upgrades, scoring 0.68. Environmental renovation assessment identified the location 2 km from township center with paved road access but requiring installation of ramps and elevators, scoring 0.62. Socioeconomic factors demonstrated strong policy support from provincial elderly care development plans with potential to employ 25 local residents, achieving the highest score of 0.84. Applying the framework's validated indicator weights through the formula $(0.75 \times 0.251) + (0.68 \times 0.268) + (0.62 \times 0.229) + (0.84 \times 0.252)$, the overall feasibility score calculated as 0.724, indicating high conversion feasibility. Based on this assessment, priority interventions were identified including structural reinforcement of load-bearing walls, installation of medical service facilities in former administrative offices, construction of barrier-free access throughout the facility, and development of partnerships with township health centers for medical support.

Policy integration and framework positioning

The ICECF framework aligns with and operationalizes key national policies including the 14th Five-Year Plan for Elderly Care Service System and the Rural Revitalization Strategy, serving as a bridge between high-level policy directives and ground-level implementation by providing specific, measurable criteria for project evaluation [20]. Local governments can integrate the framework into their elderly care facility planning processes as a standardized assessment tool on a recommended basis, while provincial authorities can use it to prioritize funding allocation across multiple conversion proposals. The framework is designed for advisory use rather than mandatory compliance, enabling flexible adaptation to local contexts and resource conditions [57].

Table 10 Framework testing protocol timeline and activities

Phase	Duration	Key activities
Pre-conversion assessment	3 months/site	Systematic application of 17 indicators; structural inspections; community surveys; cost analysis; baseline establishment
Conversion monitoring	6–12 months/site	Track framework adherence; document deviations and challenges; collect cost and timeline data
Post-conversion evaluation	12 months/site	Facility performance measurement; resident and operator surveys (n≥30); occupancy rate and service quality assessment
Framework refinement	Ongoing	Adjust indicator weights; modify assessment criteria; develop implementation guidelines

Implementation pathway and user guidelines

The framework's implementation pathway targets four primary user groups: local government planning departments who can apply it during annual elderly care facility planning cycles, project developers who can utilize it during feasibility study phases before investment decisions, policymakers who can reference it for developing conversion support policies and funding criteria, and community organizations who can employ it for advocacy and community engagement processes. The implementation process follows a structured timeline beginning with preliminary screening during the first month where local authorities identify vacant or underutilized schools using education department data, followed by framework application during months 2–3 involving systematic assessment using the 17 indicators through building inspections, community surveys, and policy analysis. Stakeholder consultation occurs in month 4 where assessment results are shared with community members, potential operators, and relevant government departments, leading to decision-making in month 5 based on framework scores and stakeholder input to prioritize projects for implementation. The final phase spans months 6–12 focusing on implementation planning where detailed conversion plans are developed for selected projects, funding is secured, and construction is initiated. This standardized approach enables comparison across multiple potential sites, facilitates evidence-based decision-making, and provides clear documentation for funding applications and regulatory approvals, with its comprehensive nature ensuring that all critical factors are considered systematically to reduce the risk of project failure due to overlooked issues.

Planned real-world testing protocol

To validate the ICECF framework through empirical application, a systematic testing protocol has been designed involving three rural schools in Henan Province preliminarily identified for pilot testing: Zhongmou County (built 1987, 3,200m^2) representing older construction, Lankao County (built 1995, 4,500m^2) representing mid-age facilities, and Luoyang area (built 2003, 5,800m^2) representing recent construction. The testing protocol encompasses four sequential phases as detailed in Table 10. The pre-conversion assessment phase lasting three months per site involves systematic application of all 17 indicators through structural inspections, community surveys, and cost analysis to establish baseline measurements. The conversion monitoring phase spanning 6–12 months per site tracks adherence to framework recommendations, documents deviations and implementation challenges, and collects cost and timeline data for each indicator category. The post-conversion evaluation phase lasting 12 months per site measures facility performance against framework predictions through surveys of residents (n≥30) and operators, assessing occupancy rates, service quality, and community acceptance while comparing predicted versus actual outcomes. The framework refinement phase synthesizes empirical findings to adjust indicator weights, modify assessment criteria, and develop detailed implementation guidelines. Initial site assessment is scheduled to begin in 2026, with phased implementation across all three sites through 2028, culminating in expected outcomes including a validation report documenting framework accuracy, refined indicator weightings based on empirical evidence, and a practical implementation handbook for future users. This planned testing protocol directly addresses the current study's limitation of hypothetical application while building upon the robust methodological foundation established through expert consensus validation.

Conclusions

Research summary

This study successfully developed and validated a comprehensive evaluation framework for the adaptive reuse of vacant rural school buildings into elderly care facilities, directly addressing a critical challenge within China's rapidly aging society. The innovative Integrated Computational-Expert Consensus Framework (ICECF) systematically identified and validated 17 key indicators, categorized across four essential dimensions: building conditions, functional conversion, environmental renovation, and socioeconomic factors.

The methodological integration of SciBERT-based natural language processing with the Fuzzy Delphi method represents a significant advance in policy research frameworks. This approach efficiently extracted evaluation indicators from extensive literature and interviews (89 academic articles, 25 policy documents, 31 stakeholder interviews), achieving strong performance metrics (F1-score: 0.85, precision: 0.87, recall: 0.83). Subsequent rigorous expert validation, involving 15 multidisciplinary

specialists, refined these indicators through consensus-building, with high consensus rates (80% to 100% for accepted indicators in the final round) for the final set.

Policy support emerged as the most critical determinant (weight: 0.0792), underscoring the imperative of governmental backing. Resource utilization (weight: 0.0669) and catering function (weight: 0.0669) also demonstrated high importance, highlighting the necessity for efficient resource management and essential service provision in successful facility conversions. The framework's four-dimensional structure provides a balanced perspective, considering both technical feasibility and community needs, alongside sustainability objectives.

Practical applications and policy recommendations

This research offers concrete solutions to the dual challenges of China's aging population and declining birth rates. While falling fertility reduces the future pool of caregivers, an aging demographic means more rural elderly may lack adequate support. Repurposing vacant rural schools into elderly care facilities directly addresses this pressing need, providing a viable strategy. The framework not only theoretically structures this assessment but, critically, offers a highly actionable tool to translate this vision into reality, effectively alleviating the scarcity of rural elderly care resources and the burden of unattended elderly.

Local governments and project developers now possess a systematic evaluation tool to guide decision-making, optimize resource allocation, and ensure comprehensive assessment of conversion projects. The framework's emphasis on both structural integrity and community integration reflects the intricate requirements for establishing elderly care facilities that are functionally adequate and socially acceptable. Furthermore, this study contributes to sustainable development by promoting the adaptive reuse of existing infrastructure, thereby reducing construction waste and preserving embodied energy. By integrating employment generation and community revitalization within its evaluation criteria, the framework ensures conversions contribute to broader rural development goals beyond direct elderly care provision.

Research contributions and future directions

This study makes two key theoretical contributions. First, it constructs a new methodological approach (ICECF) by integrating computational methods with expert knowledge, providing a systematic way to develop robust evaluation frameworks. Second, it establishes a specific evaluation framework for repurposing vacant rural school buildings into elderly care facilities, offering a structured method to address this particular societal challenge.

Future research should prioritize empirical validation through the planned three-site testing protocol detailed in the Discussion section (Planned Real-World Testing Protocol), followed by long-term outcome tracking and cross-cultural adaptation of the framework. Integrating emerging technologies like Building Information Modeling (BIM) and Artificial Intelligence (AI) could further enhance the framework's precision and accessibility [58, 59]. Facing China's increasingly acute societal aging challenges, the methodology developed in this study provides a replicable model for the systematic evaluation of infrastructure adaptation projects, poised to contribute significantly to resolving the practical challenges of rural elderly care in China.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12877-025-06772-1.

Supplementary Material 1.

Acknowledgements

The authors would like to express their sincere gratitude to the Henan Provincial Federation of Social Sciences and Henan Provincial Federation of Economic Associations for their financial support through Project SKL-2022-608. We also thank Henan Industry and Trade Vocational College and Universiti Sains Malaysia (USM) for their invaluable academic and institutional support, which greatly contributed to the successful completion of this work. The affiliation with Universiti Sains Malaysia is duly acknowledged in accordance with institutional policies and incentive schemes.

Code availability

The SciBERT model was fine-tuned based on publicly available BERT models. Specific fine-tuning parameters are described in the manuscript. Custom scripts for analysis are available from the corresponding authors upon reasonable request.

Authors' contributions

Quan Wen: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data Curation, Writing - Original Draft, Visualization. Mazran Ismail: Conceptualization, Methodology, Writing - Review & Editing, Supervision, Project administration. Muhammad Hafeez Abdul Nasir: Writing - Review & Editing, Supervision.

Funding

This work was supported by the Henan Provincial Federation of Social Sciences and Henan Provincial Federation of Economic Associations, China, Research Project "Research on Age-Friendly Design of Outer Environment Renovation in Old Residential Areas in Henan Province Based on Analytic Hierarchy Process" (Project No. SKL2022608).

Data availability

The datasets generated during this study (including anonymized interview transcripts, expert evaluation scores, SciBERT parameters and indicator lists) are available from the corresponding authors upon reasonable request at wen-quan@student.usm.my or mazran@usm.my. The semi-structured interview protocol is provided as Appendix 1: Interview Questionnaire.

Declarations

Ethics approval and consent to participate

This study received ethical approval from the Henan Industry and Trade Vocational College. All interview participants provided written informed consent. Expert participants in the Delphi study provided electronic consent.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

Received: 3 June 2025 / Accepted: 10 November 2025
Published online: 09 January 2026

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